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Ceramic heater

Abstract

A ceramic heater includes a disk-shaped ceramic plate having a wafer placement surface on a front surface thereof, the ceramic plate having a plasma electrode and a heater electrode embedded therein in this order, from closest to the wafer placement surface to farthest, in such a manner that the plasma electrode and the heater electrode are spaced apart from each other; a cylindrical shaft that supports the ceramic plate from a back surface of the ceramic plate; a plasma electrode connecting member arranged in the cylindrical shaft and connected to the plasma electrode; a heater electrode connecting member arranged in the cylindrical shaft and connected to the heater electrode; and a planar shield portion arranged on the back surface of the ceramic plate or embedded in the ceramic plate so as to be closer to the back surface than the heater electrode, the planar shield portion being connected to ground.

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Background/Summary

BACKGROUND OF THE INVENTION

1. Field of the Invention

(1) The present invention relates to a ceramic heater.

2. Description of the Related Art

(2) In the related art, a semiconductor manufacturing process may adopt a plasma CVD process. In the plasma CVD process, a wafer is placed on a wafer placement surface of a ceramic heater. A plasma electrode and a heater electrode are embedded in a ceramic plate of the ceramic heater. An upper electrode is disposed in a space above the wafer. When a high-frequency voltage is applied between the upper electrode and the plasma electrode, plasma is generated between these electrodes, and a thin film is deposited on the wafer using the plasma. In such a ceramic heater, if a leakage current produced by the high-frequency voltage applied to the plasma electrode is transmitted to the heater electrode, there is a problem that energization to the heater electrode is not controllable. In this respect, PTL 1 proposes to provide a high-resistance insulating layer (leakage-current prevention layer) between the plasma electrode and the heater electrode.

CITATION LIST

Patent Literature

(3) PTL 1: JP 3602908 B

SUMMARY OF THE INVENTION

(4) In the ceramic heater described above, however, plasma that has flowed around to a surface (back surface) of the ceramic plate opposite to the wafer placement surface may be coupled to the heater electrode. If such coupling occurs, a high-frequency current flows into the heater electrode, which is not preferable.

(5) The present invention has been made to solve such a problem, and a main object thereof is to prevent plasma that has flowed around to the back surface of a ceramic plate from being coupled to

a heater electrode.

(6) A ceramic heater of the present invention includes: a disk-shaped ceramic plate having a wafer placement surface on a front surface thereof, the ceramic plate having a plasma electrode and a heater electrode embedded therein in this order, from closest to the wafer placement surface to farthest, in such a manner that the plasma electrode and the heater electrode are spaced apart from each other; a cylindrical shaft that supports the ceramic plate from a back surface of the ceramic plate; a plasma electrode connecting member arranged in the cylindrical shaft, the plasma electrode connecting member being connected to the plasma electrode; a heater electrode connecting member arranged in the cylindrical shaft, the heater electrode connecting member being connected to the heater electrode; and a planar shield portion arranged on the back surface of the ceramic plate or embedded in the ceramic plate so as to be closer to the back surface than the heater electrode, the planar shield portion being connected to ground.

(7) In this ceramic heater, the planar shield portion prevents coupling between plasma that has flowed around to the back surface of the ceramic plate and the heater electrode. It is therefore possible to prevent the occurrence of an abnormal condition (such as abnormal heat generation or abnormal expansion of the heater electrode) caused by such coupling.

(8) In the ceramic heater of the present invention, the planar shield portion may be a conductive ceramic coating film, a metal mesh, or a metal plate. This enables the planar shield portion to be relatively easily disposed.

(9) In the ceramic heater of the present invention, the planar shield portion may be connected to the ground via a cylindrical conductor on an inner surface or an outer surface of the cylindrical shaft. This eliminates a need to allow a member to pass through the inside of the cylindrical shaft to connect the planar shield portion to the ground. Thus, other members (such as the heater electrode connecting member and the plasma electrode connecting member) are easily arranged in the space inside the cylindrical shaft. In addition, connecting the cylindrical conductor on the inner surface or the outer surface of the cylindrical shaft to the ground can prevent noise generated from the plasma from flowing to the heater electrode connecting member, the plasma electrode connecting member, and the like. Noise can be prevented even if the planar shield portion and the cylindrical conductor on the inner surface or the outer surface of the cylindrical shaft are not connected to each other.

(10) In the ceramic heater of the present invention, the planar shield portion may be parallel to the wafer placement surface. The term “parallel” includes exactly parallel as well as substantially parallel (such as within tolerance) (the same applies in the following).

(11) In the ceramic heater of the present invention, the planar shield portion may have a larger outer diameter than the heater electrode. This makes it easy to prevent coupling between the plasma and the outer peripheral portion of the heater electrode.

(12) In the ceramic heater of the present invention, the ceramic plate and the cylindrical shaft may be made of AlN ceramic, and the planar shield portion may be made of Al or an Al alloy.

(13) The ceramic heater of the present invention may include a side shield portion disposed so as to extend toward the wafer placement surface from an outer edge of the planar shield portion. Thus, the side shield portion prevents coupling between plasma that has flowed around to the side surface side of the ceramic plate and the heater electrode. It is therefore possible to prevent the occurrence of an abnormal condition (such as abnormal heat generation or abnormal expansion of the heater electrode) caused by such coupling. In this case, the side shield portion may be made of Al or an Al alloy. In this case, furthermore, the planar shield portion and the side shield portion may be embedded in the ceramic plate, and the side shield portion may have a distal end positioned between the wafer placement surface and the heater electrode. Thus, coupling between the plasma that has flowed around to the side surface side of the ceramic plate and the heater electrode can be reliably prevented.

(14) In the ceramic heater of the present invention, the planar shield portion may be embedded in the ceramic plate, and a distance from the back surface to the planar shield portion may be larger

than a distance from the wafer placement surface to the plasma electrode. The distance from the back surface to the planar shield portion affects plasma density on the wafer placement surface. Accordingly, setting the distance from the back surface to the plasma electrode to a large value can reduce the influence of the planar shield portion on the plasma on the wafer placement surface side. In addition, it is possible to easily prevent coupling between the plasma that has flowed around to the back surface side and the heater electrode. In this case, the distance from the back surface to the planar shield portion may be longer than the distance from the wafer placement surface to the plasma electrode by 2.0 mm or more. For example, when the distance from the wafer placement surface to the plasma electrode is 1.0 mm, the distance from the back surface to the planar shield portion is set to be 3.0 mm or more.

(15) In the ceramic heater of the present invention, the planar shield portion may be embedded in the ceramic plate, and the distance from the back surface to the planar shield portion may be larger than a distance from the planar shield portion to the heater electrode. Thus, the plasma that has flowed around from the back surface can be more effectively prevented by the planar shield portion positioned near the heater electrode.

(16) In the ceramic heater of the present invention, the planar shield portion may be arranged on the back surface of the ceramic plate, the side shield portion may be arranged on a side surface of the ceramic plate, and the side shield portion may have a distal end positioned closer to the back surface than a plane on which the heater electrode lies. This makes it easy to maintain thermal uniformity across the wafer placement surface. In this case, the planar shield portion may be arranged on the back surface of the ceramic plate with a low thermal conductive layer interposed therebetween, the low thermal conductive layer having lower thermal conductivity than the ceramic plate. This allows the low thermal conductive layer to prevent the heat of the ceramic plate from being directly transmitted to the planar shield portion, and thus the planar shield portion can be prevented from being softened or melted.

(17) In the ceramic heater of the present invention, preferably, the planar shield portion is not disposed between the plasma electrode and the heater electrode. The high-frequency power applied to the plasma electrode affects plasma density on the wafer placement surface if the high-frequency power flows to another conductor and is wastefully consumed. For this reason, the planar shield portion is preferably disposed between the back surface and the heater electrode so as to prevent the high-frequency power from flowing from the plasma electrode to the planar shield portion.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a longitudinal sectional view of a plasma treatment device 10.
- (2) FIG. 2 is a longitudinal sectional view of a modification of the plasma treatment device 10.
- (3) FIG. 3 is a longitudinal sectional view of a modification of the plasma treatment device 10.
- (4) FIG. 4 is a longitudinal sectional view of a modification of the plasma treatment device 10.
- (5) FIG. 5 is a longitudinal sectional view of a modification of the plasma treatment device 10.
- (6) FIG. 6 is a longitudinal sectional view of a modification of the plasma treatment device 10.

DETAILED DESCRIPTION OF THE INVENTION

(7) A preferred embodiment of the present invention will be described hereinafter with reference to the drawings. FIG. 1 is a longitudinal sectional view of a plasma treatment device 10. In FIG. 1, dotted-line arrows indicate plasma or plasma current.

(8) As illustrated in FIG. 1, the plasma treatment device 10 includes a chamber 12, an upper electrode 14, and a ceramic heater 20.

(9) The chamber 12 is a container made of an aluminum alloy or the like and formed into a box shape. The chamber 12 is provided with a pressure regulating valve, a vacuum pump, or the like

(not illustrated) such that the inside of the chamber **12** can be adjusted to a desired pressure.

(10) The upper electrode **14** is attached to the ceiling of the chamber **12**. The upper electrode **14** is connected to an RF power supply **16** for plasma generation.

(11) The ceramic heater **20** includes a ceramic plate **21**, a cylindrical shaft **26**, and a conductive film **30**.

(12) The ceramic plate **21** is a disk-shaped AlN ceramic member having a wafer placement surface **21a** on a front surface thereof. The ceramic plate **21** has a plasma electrode **22** and a heater electrode **24** embedded therein in this order, from closest to the wafer placement surface **21a** to farthest, in such a manner that the plasma electrode **22** and the heater electrode **24** are spaced apart from each other.

(13) The cylindrical shaft **26** supports the ceramic plate **21** from a back surface **21b** of the ceramic plate **21**. The cylindrical shaft **26** is mounted on a perforated metal part **28**. The inside of the cylindrical shaft **26** and the hole in the metal part **28** communicate with each other. The metal part **28** is connected to ground. The cylindrical shaft **26** has arranged therein a plasma electrode connecting member **23** connected to the plasma electrode **22**, and a heater electrode connecting member **25** connected to the heater electrode **24**. The plasma electrode connecting member **23** is connected to the ground. The heater electrode connecting member **25** applies a voltage across the heater electrode **24** to cause the heater electrode **24** to generate heat.

(14) The conductive film (corresponding to the planar shield portion of the present invention) **30** is disposed in an annular region on the back surface **21b** of the ceramic plate **21**, except for a portion corresponding to the cylindrical shaft **26**. The conductive film **30** is preferably parallel to the wafer placement surface **21a**. The conductive film **30** may be a coating film made of conductive ceramic. Examples of the conductive ceramic include SiC, TiC, and TiN. The conductive film **30** is connected to the metal part **28** (ground) via a conductive wire **31** disposed over the outer wall of the cylindrical shaft **26** so as to extend in the up-down direction. The conductive wire **31** may be a linear coating film made of conductive ceramic.

(15) In the ceramic heater **20** of this embodiment described above, the plasma current acting from the back surface side of the ceramic plate **21** flows to the ground via the conductive film **30**. That is, the conductive film **30** prevents coupling between the plasma that has flowed around to the back surface side of the ceramic plate **21** and the heater electrode **24**. It is therefore possible to prevent the occurrence of an abnormal condition (such as abnormal heat generation or abnormal expansion of the heater electrode **24**) caused by such coupling.

(16) If the conductive wire **31** is a linear coating film made of conductive ceramic disposed over the outer wall of the cylindrical shaft **26** so as to extend in the up-down direction, wiring of a conductive wire inside the cylindrical shaft **26** is not necessary. In addition, the influence of the conductive wire **31** on the chamber side is smaller than that in a case where a conductive wire is disposed in the chamber **12**.

(17) It is understood that the present invention is not limited to the embodiment described above in any way and may be carried out in various modes so long as they fall within the technical scope of the present invention.

(18) For example, a metal cover **40** and a conductive wire **41** illustrated in FIG. 2 may be used instead of the conductive film **30** and the conductive wire **31** in the embodiment described above. In FIG. 2, the same components as those in the embodiment described above are denoted by the same numerals. The metal cover **40** (corresponding to the planar shield portion of the present invention) is a doughnut-shaped metal plate and is disposed in an annular region on the back surface **21b** of the ceramic plate **21**, except for a portion corresponding to the cylindrical shaft **26**. The metal cover **40** is secured to the back surface **21b** of the ceramic plate **21** with a low thermal conductive layer **42** interposed therebetween, the low thermal conductive layer **42** having lower thermal conductivity than the ceramic plate **21**, by screws **43**. The metal cover **40** is preferably parallel to the wafer placement surface **21a**. The low thermal conductive layer **42** is made of, for

example, alumina ceramic when the ceramic plate **21** is made of AlN ceramic. The metal cover **40** is connected to the metal part **28** (ground) via the conductive wire **41** provided outside the cylindrical shaft **26**. In FIG. 2, the metal cover **40** prevents coupling between the plasma that has flowed around to the back surface side of the ceramic plate **21** and the heater electrode **24**. It is therefore possible to prevent the occurrence of an abnormal condition (such as abnormal heat generation or abnormal expansion of the heater electrode **24**) caused by such coupling. In addition, since the low thermal conductive layer **42** prevents the heat of the ceramic plate **21** from being directly transmitted to the metal cover **40**, the metal cover **40** can be prevented from being softened or melted. Such an effect is effectively obtained, in particular, when the metal cover **40** is made of Al or an Al alloy and the low thermal conductive layer **42** is an alumina layer. Like the conductive wire **31** in FIG. 1, the conductive wire **41** in FIG. 2 may be disposed over the outer wall of the cylindrical shaft **26** so as to extend in the up-down direction.

(19) Alternatively, a shield electrode **50** and a conductive wire **51** illustrated in FIG. 3 may be used instead of the conductive film **30** and the conductive wire **31** in the embodiment described above. In FIG. 3, the same components as those in the embodiment described above are denoted by the same numerals. The shield electrode **50** (corresponding to the planar shield portion of the present invention) is a metal planar member embedded in the ceramic plate **21** so as to be closer to the back surface side than the heater electrode **24**. The shield electrode **50** is preferably parallel to the wafer placement surface **21a**. The shield electrode **50** may be a mesh of a metal material (such as Mo). The shield electrode **50** is connected to the metal part **28** (ground) via the conductive wire **51** disposed inside the cylindrical shaft **26**. In FIG. 3, the shield electrode **50** prevents coupling between the plasma that has flowed around to the back surface side of the ceramic plate **21** and the heater electrode **24**. It is therefore possible to prevent the occurrence of an abnormal condition (such as abnormal heat generation or abnormal expansion of the heater electrode) caused by such coupling.

(20) A shield electrode **54** illustrated in FIG. 4 may be used instead of the shield electrode **50** in FIG. 3. The shield electrode **54** includes a planar shield portion **52** and a side shield portion **53**. The planar shield portion **52** is a metal planar member embedded in the ceramic plate **21** so as to be closer to the back surface **21b** than the heater electrode **24** and having a larger outer diameter than the heater electrode **24**. The planar shield portion **52** is parallel to the wafer placement surface **21a**. A distance D2 from the back surface **21b** to the planar shield portion **52** is preferably larger than a distance D1 from the wafer placement surface **21a** to the plasma electrode **22**. More preferably, the distance D2 is larger than the distance D1 by 2.0 mm or more. The distance D2 is larger than a distance D3 from the planar shield portion **52** to the heater electrode **24**. The planar shield portion **52** is connected to the metal part **28** (ground) via the conductive wire **51** disposed inside the cylindrical shaft **26**. The side shield portion **53** is a metal cylindrical member disposed so as to extend toward the wafer placement surface **21a** from the outer edge of the planar shield portion **52**. The side shield portion **53** has a distal end positioned between the wafer placement surface **21a** and the heater electrode **24** (specifically, between the plasma electrode **22** and the heater electrode **24**). The planar shield portion **52** and the side shield portion **53** may be a mesh of a metal material (such as Mo). In FIG. 4, the planar shield portion **52** prevents coupling between the plasma that has flowed around to the back surface side of the ceramic plate **21** and the heater electrode **24**. It is therefore possible to prevent the occurrence of an abnormal condition (such as abnormal heat generation or abnormal expansion of the heater electrode) caused by such coupling. In addition, the planar shield portion **52** has a larger outer diameter than the heater electrode **24**. This makes it easy to prevent coupling between the plasma and the outer peripheral portion of the heater electrode **24**. Furthermore, the side shield portion **53** prevents coupling between the plasma that has flowed around to the side surface side of the ceramic plate **21** and the heater electrode **24**. It is therefore possible to prevent the occurrence of an abnormal condition caused by such coupling. The distal end of the side shield portion **53** is positioned between the wafer placement surface **21a** and the

heater electrode **24**. Thus, coupling between the plasma that has flowed around to the side surface side of the ceramic plate **21** and the heater electrode **24** can be reliably prevented. Furthermore, the distance D2 from the back surface **21b** to the planar shield portion **52** is larger than the distance D1 from the wafer placement surface **21a** to the plasma electrode **22**. This can reduce the influence of the planar shield portion **52** on the plasma on the wafer side. Furthermore, the distance D2 from the back surface **21b** to the planar shield portion **52** is larger than the distance D3 from the planar shield portion **52** to the heater electrode **24**. This makes it less likely that the plasma flows into the planar shield portion **52** from the back surface **21b**, and can reduce the influence of the planar shield portion **52** on the plasma on the wafer placement surface side.

(21) A conductor **56** illustrated in FIG. 5 may be used instead of the conductive wire **51** in FIG. 4. The conductor **56** is disposed so as to cover the entire inner surface of the cylindrical shaft **26**. As illustrated in FIG. 5, the planar shield portion **52** is connected to the conductor **56** via a conductive wire **61** embedded in the ceramic plate **21**, and the conductor **56** is connected to the metal part **28** (ground). The conductor **56** may be a coating film made of conductive ceramic. Examples of the conductive ceramic include SiC, TiC, and TiN. In FIG. 5, it is not necessary to allow a member to pass through the inside of the cylindrical shaft **26** to connect the shield electrode **54** to the metal part **28**. Thus, other members (such as a heater electrode connecting member and a plasma electrode connecting member) are easily arranged in the space inside the cylindrical shaft **26**. In FIG. 5, the conductor **56** is disposed on the inner surface of the cylindrical shaft **26**. However, the conductor **56** may be disposed on the outer surface of the cylindrical shaft **26**. While the conductor **56** is disposed on the entire inner surface of the cylindrical shaft **26**, the conductor **56** may be disposed in a line from the upper end to the lower end of the cylindrical shaft **26** (to a position adjoining the metal part **28**), rather than disposed on the entire surface.

(22) A metal structure **70** illustrated in FIG. 6 may be used instead of the metal cover **40** and the conductive wire **41** in FIG. 2. The metal structure **70** is formed by integrating a metal sleeve **71** and a metal cover **74**. The metal sleeve **71** covers the entire outer surface of the cylindrical shaft **26**. The metal sleeve **71** is secured to the metal part **28** by conductive bolts **55** through a flange **27** at the lower end of the cylindrical shaft **26**. Accordingly, the metal structure **70** is connected to the metal part **28** (ground) via the bolts **55**. The metal cover **74** includes a planar shield portion **72**, which is a doughnut-shaped metal plate, and a side shield portion **73** disposed so as to extend upward from the outer edge of the planar shield portion **72**. The planar shield portion **72** is arranged on the back surface **21b** with the low thermal conductive layer **42** interposed therebetween, is parallel to the wafer placement surface **21a**, and has a larger outer diameter than the heater electrode **24**. The side shield portion **73** has a distal end positioned between a plane on which the heater electrode **24** lies and the back surface **21b**. To attach the metal structure **70** to the cylindrical shaft **26**, for example, the metal structure **70** is vertically halved into two half members, and the two half members are arranged so as to surround the cylindrical shaft **26**, and are then secured to the cylindrical shaft **26** by a fastening tool **75** (such as bolts or a cable tie). In FIG. 6, the planar shield portion **72** prevents coupling between the plasma that has flowed around to the back surface side of the ceramic plate **21** and the heater electrode **24**. In addition, the planar shield portion **72** has a larger outer diameter than the heater electrode **24**. This makes it easy to prevent coupling between the plasma and the outer peripheral portion of the heater electrode **24**. Furthermore, the side shield portion **73** prevents coupling between the plasma that has flowed around to the side surface side of the ceramic plate **21** and the heater electrode **24**. It is therefore possible to prevent the occurrence of an abnormal condition caused by such coupling. The distal end of the side shield portion **73** is positioned between a plane on which the heater electrode **24** lies and the back surface **21b**. This makes it easy to maintain thermal uniformity across the wafer placement surface **21a**. Moreover, since the low thermal conductive layer **42** prevents the heat of the ceramic plate **21** from being directly transmitted to the metal cover **74**, the metal cover **74** can be prevented from being softened or melted. Such an effect is effectively obtained, in particular, when the metal cover **74** is

made of Al or an Al alloy and the low thermal conductive layer **42** is an alumina layer. Since the metal cover **74** is supported by the integrated metal sleeve **71**, the screws **43** in FIG. **2** can be omitted. Omission of the screws **43** makes it easy to maintain thermal uniformity across the ceramic plate **21**. The metal cover **40** in FIG. **2** may have a side shield portion. Alternatively, a conductive film or a metal sleeve that covers the entire outer surface of the cylindrical shaft **26** may be used instead of the conductive wire **31** in FIG. **1** or the conductive wire **41** in FIG. **2**.

(23) In the embodiment described above, the upper electrode **14** is connected to the RF power supply **16**, and the plasma electrode **22** is connected to the ground. Alternatively, the plasma electrode **22** may be connected to the RF power supply **16**, and the upper electrode **14** may be connected to the ground. The same applies to FIG. **2** to FIG. **6**.

(24) The present application claims priority from JP Patent Application No. 2019-219545 filed Dec. 4, 2019, the entire contents of which are incorporated herein by reference.

Claims

1. A ceramic heater comprising: a disk-shaped ceramic plate having a wafer placement surface on a front surface thereof, the ceramic plate having a plasma electrode and a heater electrode embedded therein in this order, from closest to the wafer placement surface to farthest, in such a manner that the plasma electrode and the heater electrode are spaced apart from each other; a cylindrical shaft that supports the ceramic plate from a back surface of the ceramic plate; a plasma electrode connecting member arranged in the cylindrical shaft, the plasma electrode connecting member being connected to the plasma electrode; a heater electrode connecting member arranged in the cylindrical shaft, the heater electrode connecting member being connected to the heater electrode; and a planar shield portion arranged on the back surface of the ceramic plate or embedded in the ceramic plate so as to be closer to the back surface than the heater electrode, the planar shield portion being connected to ground; and a side shield portion disposed so as to extend toward the wafer placement surface from an outer edge of the planar shield portion, wherein the planar shield portion and the side shield portion are embedded in the ceramic plate, and the side shield portion has a distal end positioned between the wafer placement surface and the heater electrode.
 2. The ceramic heater according to claim 1, wherein the planar shield portion is a conductive ceramic coating film, a metal mesh, or a metal plate.
 3. The ceramic heater according to claim 1, wherein the planar shield portion is connected to the ground via a cylindrical conductor on an inner surface or an outer surface of the cylindrical shaft.
 4. The ceramic heater according to claim 1, wherein the planar shield portion has a larger outer diameter than the heater electrode.
 5. The ceramic heater according to claim 1, wherein the ceramic plate and the cylindrical shaft are made of AlN ceramic, and the planar shield portion is made of Al or an Al alloy.
 6. The ceramic heater according to claim 1, wherein the side shield portion is made of Al or an Al alloy.
 7. The ceramic heater according to claim 1, wherein a distance from the back surface to the planar shield portion is larger than a distance from the wafer placement surface to the plasma electrode.
 8. The ceramic heater according to claim 7, wherein the distance from the back surface to the planar shield portion is longer than the distance from the wafer placement surface to the plasma electrode by 2.0 mm or more.
 9. The ceramic heater according to claim 1, wherein the distal end of the side shield portion does not reach the height of the plasma electrode.
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